INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, MECHANICAL, STORAGE STAND (FOR BULK MILK DISPENSERS) (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers refrigerators for bulk milk containers, which also serve as counter-height stands for supporting bulk milk dispensers for Naval shipboard use.

2. CLASSIFICATION

Size 1: Storage capacity for two 5-gallon milk containers
Size 2: Storage capacity for three 5-gallon milk containers

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The refrigerator shall consist of an insulated storage compartment having a front-mounted insulated door and a top section to rigidly support a fully-loaded two or three 5-gallon container bulk milk dispenser. The refrigerator shall simultaneously support, in the storage compartment, a full complement of bulk milk containers. The refrigerator shall accommodate the size, weight, and loading impact of fully-loaded, 5-gallon, dispenser-type milk containers. The refrigerator shall also be suitable for use with plastic, single service and multiple service bulk milk containers. The size 1 refrigerator shall support a two 5-gallon container dispenser; the size 2 refrigerator shall support a three 5-gallon container dispenser. The refrigerator shall consist of a complete refrigeration system with automatic controls necessary to maintain the operating temperatures specified. The storage compartment shall consist of an outer shell, inner liner, framing, door, hardware, and leg mounting. The outer shell and inner liner shall enclose insulation.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson-Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 4110

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3.2 Dimensions. Refrigerator dimensions shall be within the limits identified in Table 1.

Table 1. Dimensions (inches).

Refrigerator Size	Acceptable range of exterior dimensions			Minimum interior dimensions		
	Height	Width	Depth	Height	Width	Depth
1	34-36	27-30	24-25	22	22	11
2	34-36	38-40	24-25	22	32	11

3.2.1 Physical size limitations. The unit shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8 inch radius corners. When specified for submarines, the cabinet shall pass or shall disassemble so that it can pass through a 30 inch diameter submarine access. The refrigeration system shall not require disassembly to pass through access openings. (Shipboard requirement, see 7.2)

3.3 Performance. The refrigerators shall meet the following performance requirements:

a. In an ambient air temperature surrounding the unit of 100°F, the refrigerator shall be capable of lowering the temperature of the empty storage compartment from 100°F to a stabilized temperature of 40 ± 2°F in not greater than 60 minutes.

b. The unit, when loaded to capacity with fully-loaded bulk milk containers, shall, after stabilization, be capable of maintaining the compartment temperature at 40 ± 2°F in an ambient air test temperature surrounding the unit of 100°F or any constant or fluctuating temperature between 50 and 100°F.

c. With the compartment temperature stabilized at 40 ± 2°F in an ambient air test temperature surrounding the unit of 100°F, the maximum operating (on) time of the compressor under no-load conditions shall be not greater than 75 percent of a continuous 3-hour period.

3.4 Outer shell. The outer shell for the storage compartment shall be stainless steel class 302 or 304 in accordance with ASTM A 167 or class 430 in accordance with ASTM A 176. Where required, panels or covers shall be removable and shall provide ready access for servicing of the refrigeration system.

3.5 Inner liner. The thickness of the walls and top of the inner liner shall be not less than 24-gauge stainless steel. The thickness of the liner bottom shall be not less than 18-gauge stainless steel. The liner bottom shall be formed and braced to support a full complement of fully-loaded bulk milk cans and to withstand, without permanent deflection or distortion, the impact normally associated with the loading operation. The liner bottom shall be connected to the liner walls so that no liquid can leak into the insulation. This connection shall be able to withstand the impact of the loading operation.

3.6 Framing. Framing shall support the fully-loaded dispenser and the fully-loaded storage compartment, and to house the components of the refrigeration system. The thickness of the framing shall be not less than 12-gauge structural steel shapes or plates. The steel framing shall be painted or treated to resist corrosion.

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3.7 Door. The door shall consist of an inner and outer liner of the same material and thickness specified for the cabinet liner and shell. The door shall be fitted with a gasket. The door framing shall include suitable plastic or synthetic rubber breaker strips to minimize conduction between the cabinet liner and shell. The door shall be hinged on the right side unless a left hand door is specified. Door hinges and framing shall be of sufficient strength to prevent misalignment of the door during opening or closing. A thermometer shall be mounted on the door.

3.8 Hardware. The door latch, striker plate, and hinges shall be stainless steel and shall be the same quality used for the intended purpose in commercial practice.

3.9 Mounting. The refrigerator shall be mounted on four 6-inch high stainless steel legs suitable for secure attachment to the deck.

3.10 Refrigeration system. The refrigeration system shall operate by static or forced convection, and shall consist of a hermetic motor-compressor unit, an air cooled condenser, a drier, evaporator sections with a capillary tube or thermal expansion valve, and an automatic thermostat control. The refrigeration system shall be completely dehydrated and fully charged prior to shipment.

3.10.1 Refrigerant (ozone depleting substance). The refrigerant shall be nonflammable, be in accordance with ARI 700, and be approved by the Environmental Protection Agency under the Significant New Alternate Program. (It shall have an ozone depletion potential of not greater than 0.05.)

3.11 Electrical requirements. The unit shall operate on 115 VAC, 60 Hz, single-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 7.2)

3.12 Motor. For size 1, the motor shall be not less than 1/8 horsepower. For size 2, the motor shall be not less than 1/6 horsepower. A low pressure sensing switch shall be provided. In the event that loss of refrigerant makes the charge sufficiently low, the unit shall automatically shut off. The low pressure sensing switch shall be the manual-reset type.

3.13 Inclined operation. The unit shall operate satisfactorily, in accordance with the requirements of this CID (such as door remaining closed to prevent loss of contents), when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

3.14 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with

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MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.15 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 7 and UL 471.

3.16 Finish. Stainless steel shall have a type 3 or smoother finish, in accordance with ASTM A 480.

3.17 Label plates. The unit shall be provided with a data nameplate and an instruction plate. They shall be attached to the unit, readily visible during normal operating use, and shall not adversely affect the life and utility of the unit.

3.17.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.17.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Size required
- When required, left hand door swing
- When required, manuals shall be in accordance with ASTM F 760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

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7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)
MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating

7.3.2 Air-Conditioning & Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700 - Specifications for Fluorocarbons and Other Refrigerants

7.3.3 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip
ASTM A 480 - General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM F 760 - Food Service Equipment Manuals

7.3.4 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 7 - Food Service Refrigerators and Storage Freezers

7.3.5 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 471 - Commercial Refrigerators and Freezers

7.3.6 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hose Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.3.7 The National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

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Norris Dispenser Company
13430 County Road No. 6
Plymouth, MN 55441-4571

Silver King
Division of Stevens Lee Co.
1600 Xenium Ln, N.
Minneapolis, MN 55441-3706

8. CONCLUDING MATERIAL

Custodian:
  Navy - SH

Preparing activity:
 Navy - SH
 (Project 4410-0525)

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